Jeanne Boursier
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CURRICULUM VITAE

APPOINTMENTS

2026–present	Assistant Professor (tenure track), MIT, Cambridge
2023–2026	Ritt Assistant Professor (postdoctoral position), Columbia University, New York
2022–2023	Postdoctoral researcher, ENS Lyon (UMPA), ERC Project LDRAM

EDUCATION

2019–2022	PhD in Mathematics, Université Paris-Dauphine. Advisor: Djalil Chafaï; co-advisor: Sylvia Serfaty (New York University)
2015–2019	Élève normalien (supported student), École Normale Supérieure Paris-Saclay, Cachan

DEGREES

2018	MSc in Mathematics (Probability), Université Pierre et Marie Curie — highest honors
2016	BSc in Mathematics, École Normale Supérieure Paris-Saclay, Cachan

GRANTS AND DISTINCTIONS

2023	PhD Thesis Award "Prix de Thèse Jacques Neveu 2023"
2021–2024	Member of the ANR project GaLoPée, "Portrait de phase des gaz avec interactions à longue portée"
2019	Three-year PhD fellowship, Fondation CFM pour la Recherche (awarded annually to two students in Mathematics)

TEACHING

Spring 2026	Ordinary Differential Equations, Columbia University
Fall 2025	Introduction to Modern Analysis II, Columbia University
Spring 2025	Honors Math B, Columbia University
Fall 2024	Ordinary Differential Equations, Columbia University
Spring 2024	Calculus 3, Columbia University
Fall 2023	Ordinary Differential Equations, Columbia University
2021–2022	Teaching assistant in Analysis, Université Paris-Dauphine
2020–2021	Teaching assistant in Optimization (Master's level), Université Paris-Dauphine
2019–2021	Teaching assistant in Probability, Université Paris-Dauphine
2016–2018	Oral examiner in Mathematics, Lycée Saint Louis, Paris

MENTORING

2021	Co-supervisor (with Clément Berenfeld) of three Master's students in Mathematics at Université Paris-Dauphine; thesis: "Neural networks as interacting particle systems"

INVITED TALKS AND SEMINARS

October 2026	Joint probability seminar, Lehigh University and University of Minnesota (virtual)
September 2026	Probability seminar, MIT
April 2026	Probability seminar, Duke University
April 2026	Probability seminar, University of Chicago
February 2026	Probability seminar, Cornell University
December 2025	Applied Mathematics colloquium, Brown University
December 2025	Mathematics colloquium, Indiana University, Bloomington
December 2025	Mini-course, conference "Topological phase transitions and localization of random fields," University of Chile
April 2025	Probability seminar, NYU Courant
November 2024	Probability seminar, Princeton University
October 2024	Joint probability seminar, Temple University and University of Pennsylvania
October 2024	Probability seminar, CRM–ISM, Montreal
September 2024	Probability seminar, Columbia University
August 2024	Bernoulli–IMS Congress in Probability and Statistics, session on random matrices and applications
March 2023	Probability seminar, Université Aix-Marseille
February 2023	Conference "Hyperuniform structures, rigid point processes and related topics," Université de Lille
January 2023	Probability seminar, Université Grenoble Alpes
January 2023	Working group in stochastic modeling, LPSM, Sorbonne Université
December 2022	Conference "Coulomb gases and universality," Institut Henri Poincaré, Paris
November 2022	Workshop Journées EFI, Université Lyon 1
September 2022	Working group LDRAM, ENS Lyon
May 2022	Probability seminar, Université Rennes 1
April 2022	Séminaire MEGA, Institut Henri Poincaré
April 2022	Probability seminar, Université Paris-Saclay
March 2022	Probability seminar "Les probas du vendredi," LPSM, Sorbonne Université
January 2022	Working group in probability, Université de Lille
May 2021	SMAI young researchers project (BOUM, point processes), Université de Paris

ARTICLES AND PREPRINTS

1. J. Boursier, D. Chafaï, C. Labbé. *Universal cutoff for the Dyson Ornstein–Uhlenbeck process.* Probability Theory and Related Fields **185** (2023), no. 1, 449–512.

2. J. Boursier. *Optimal local laws and CLT for the circular Riesz gas.* arXiv:2112.05881 (2021). Lean 4 formalization available on GitHub.

3. J. Boursier. *Decay of correlations and thermodynamic limit for the circular Riesz gas.* arXiv:2209.00396 (2022). Lean 4 formalization available on GitHub.

4. J. Boursier, A. Guionnet. *Large deviations for the smallest eigenvalue of a deformed GOE with an outlier.* arXiv:2408.09256 (2024).

5. J. Boursier, S. Serfaty. *Dipole formation in the two-component plasma.* arXiv:2410.01025 (2024). Lean 4 formalization available on GitHub.

6. J. Boursier, S. Serfaty. *Multipole and Berezinskii–Kosterlitz–Thouless transitions in the two-component plasma.* arXiv:2509.09449 (2025). Lean 4 formalization available on GitHub.

7. J. Boursier. *The Legendre structure of the TAP complexity for the Ising spin glass.* arXiv:2604.20660 (2026).

8. J. Boursier. *Large deviations for the maximum of the generalized TAP free energy.* arXiv:2607.29330 (2026).

9. J. Boursier, J.-C. Mourrat. *Free energy of non-convex multi-species Ising spin glasses with external fields.* arXiv:2610.18350 (2026).

10. A. Ali, J. Boursier, H.-B. Chen, F.-H. Ho, V. Issa, J.-C. Mourrat. *Free energy of the Hopfield model.* arXiv:2611.03306 (2026).

11. J. Boursier, T. Leblé. *Existence of a fluid phase for the two-dimensional one-component plasma.* arXiv:2611.17174 (2026).